Kaustubh Hemant Sonawane

Mumbai, India | ksstubh.sonawane@gmail.com | +91 8591441019 | www.github.com/Arishtanemi4 |
www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

Lead AI / ML Engineer | GenAI / LLM Engineer | AI Solutions Engineer | Forward Deployed Engineer | Machine Learning Engineer | MLOps Engineer

PROFILE SUMMARY

AI / ML Engineer and de facto team lead with 3 years of industry experience owning the full lifecycle of production **GenAI and LLM applications**: architecture, **RAG**, agentic tool calling, **Azure OpenAI**, self-hosted **Llama-3**, **LlamaIndex**, **pgvector**, **FastAPI**, **ReactJS** and **AWS** deployment with **Docker** and **CI/CD**. Guided up to 4 interns and worked directly with Compliance, Client Service and Investment stakeholders. Recently completed an MSc in Data Science at the University of Bristol (UK), awaiting result. Seeking Lead AI / ML Engineer roles in India.

KEY SKILLS

- **AI / ML:** Generative AI, LLM applications and agents, RAG, Azure OpenAI, LlamaIndex, Llama-3, embeddings, pgvector, tool / function calling, prompt engineering, NLP, computer vision and deep learning (CNN, ViT, ResNet), scikit-learn.
- **Languages:** Python (pandas, FastAPI, uvicorn, scikit-learn, fuzzywuzzy), R (dplyr, tidyverse, ggplot2), SQL, NodeJS (Express, ReactJS), Shell.
- **Cloud / MLOps:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, Textract, Security Groups, CloudFormation templates), Docker, Jenkins, GitLab CI/CD, Nginx, Linux, LDAP (RBAC).
- **Databases and Tools:** PostgreSQL (schema design, functions and procedures, pgvector), SQLite, Power BI, Jupyter, Anaconda, Git, VS Code, Claude Code.

WORK EXPERIENCE

HDFC Asset Management Company Ltd. **July 2022 – July 2025**

AI Solutions Engineer / Forward Deployed Engineer

GenAI / LLM / Retrieval Augmented Generation (RAG) Engineer

- Owned the GenAI applications from architecture to release, building them end to end (**ReactJS**, **FastAPI**, **RAG** logic, **vector database**, **embeddings**) with **LlamaIndex**, **Azure OpenAI GPT-3.5** and **Llama-3** for Compliance, Client Service and Investment teams.
- Maintained department-wise **vector databases** (**pgvector**, **Azure OpenAI embeddings**) for **semantic search**; allocated models by cost, with **Llama-3 self-hosted on a GPU EC2 instance** for Client Service and **Azure OpenAI** for other departments.
- Built AI agents that call Python functions as tools to run fund-comparison calculations and plot PDF reports on request, guardrailed with **system prompts** and **prompt versioning in Git**.
- Developed a **Python** automation job (**Azure OpenAI**, earlier **AWS Textract**, with **fuzzywuzzy** matching) on **AWS SageMaker**, scheduled via **EventBridge** and **Lambda**, to parse **300+ daily emails** from credit rating agencies into a daily consolidated analyst report, **saving 60 manual analyst hours a month**.

Technical Leadership and Forward Deployed Engineering

- Worked directly with Client Service, Compliance and Investment stakeholders; the only engineer alongside the manager in year 1, later a **de facto team lead** guiding up to 4 interns as the team grew.
- Designed the platform as modular microservices: the **ReactJS** frontend served with **Nginx** on one **AWS EC2** instance and the **FastAPI** services (**Docker**, uvicorn) on another, with department-based access and **RBAC** through **LDAP** authentication.

MLOps / Cloud Infrastructure

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** utilizing **EC2, S3, ECR and Lambda**, provisioning EC2 instances with security-team **CloudFormation** templates.
- Restricted app access to company private IPs with **AWS Security Groups** and deployed an **Nginx reverse proxy**, one of the fixes that took the app from failing to passing the security team's **VAPT**.
- Built the deployment automation with **Jenkins** and **GitLab** (CI/CD pipeline): a one-click deploy of the dev or prod branch, with separate frontend and backend repositories, each in its own **Docker** container.

Data Engineer

- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) and developed **Optimized SQL** queries, **SQL Functions** and **Procedures** for preprocessing and efficient data insertion.
- Wrote **Python** ingestion scripts for the enterprise-licensed **Morningstar API**, refreshing data fortnightly (fixed income) and monthly (equity) for every fund in the Indian mutual fund market (2023–2025), and integrated it with legacy Excel data.

Data Analyst

- Developed **Power BI** dashboards for the investment team's fortnightly credit report analysis, moving analysts from Excel and saving **10 hours a week**, then recreated them in **ReactJS** for the wider organization at lower cost.
- Analyzed customer data with **Python Pandas** and identified buying patterns using **Market Basket Analysis** (Apriori Algorithm) for the marketing team.

PROJECTS

- **University of Bristol MSc Dissertation with AstraZeneca:** CellLineSelector (team project; ranks cancer cell lines from target inclusion and exclusion genes. Personal focus: preprocessing, scoring algorithm and RAG layer.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** LLM application (108 AI personas across 12 archetypes; Pydantic-validated JSON output; vision-model photo scoring. React, FastAPI.) | GitHub: https://github.com/Arishtanemi4/wingman.git
- **MSc Applied Data Science Group Project:** VisionCaptioner (team project; CNN/ViT/ResNet encoders vs LSTM/attention-based decoders for image captioning, benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator (car trained using a CNN model).

ACHIEVEMENTS

- **HDFC AMC Best Team Project 2024 and 2025 (internal recognition):** rating-change email automation (2024) and department-specific RAG chatbot (2025).
- **Project Hack 27:** Rolls Royce Derby 4th Place winner. (Human centric data analytics tool.)
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Spec-driven AI dating app.)
- **University of Bristol Datathon (Lloyd's Bank):** 3rd Place winner. (Credit card fraud detection.)

EDUCATION

University of Bristol, UK – MSc. Data Science (awaiting result) — **September 2025 – September 2026**
NMIMS, Mumbai – B.Tech IT (Dual Degree, CGPA 3.27/4) — **June 2017 – April 2022**
NMIMS, Mumbai – MBA Technology Management (Dual Degree) — **June 2017 – April 2022**

CERTIFICATIONS

- **Udemy:** Practical Deep Learning (Intel); Applied Data Science with Python; Mathematical Foundation for ML and AI.
- **Claude Code in Action:** https://verify.skilljar.com/c/4rride9vnwmv